Programming Language 1
Python

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USER INPUTS IN PYTHON

- Most programs are written to solve end user’s problem
- We need to get input from the users
- Check whether a person can vote or not ?
- Check whether a number is even or odd ?

HOW TO ACCEPT NUMERICAL INPUTS ?

```
number = input('Enter a number: ')

print(number)

print(type(number))
```

HOW TO ACCEPT NUMERICAL INPUTS?

```
number = input('Enter a number: ')

number = int(number)

print(number)

print(type(number))
```

HOW TO ACCEPT A LIST?

```
numbers = input('Enter the numbers separated by a space: ')

print(numbers)

print(type(numbers))
```

HOW TO ACCEPT A LIST?

```
numbers = input('Enter the numbers separated by a space: ')

numbers = list(numbers)

print(numbers)

print(type(numbers))
```

HOW TO REMOVE SPACE FROM A LIST?

```
numbers = input('Enter the numbers separated by a space: ')

numbers = numbers.split(' ')

print(numbers)

print(type(numbers))
```

```
numbers = input('Enter the numbers separated by a space: ')

numbers = numbers.split(' ')

list_length = len(numbers)

for i in range(list_length):

    numbers[i] = int(numbers[i])

print(numbers)

print(type(numbers))
```

PROGRAMS TO PRACTICE

- Check if a number is odd or even
- Check if one can vote or not
- Find the sum of all the numbers

Assignment:

- Print the table of a number (take input from user)
- Check if a number is multiple of 10 or not (take input from the user)

WHILE LOOP IN PYTHON

WHILE LOOP IN PYTHON

Counting number from 1 to 5

```python
current_number = 1

while current_number <= 5:

    print(current_number)

    current_number += 1
```

LETTING THE USER CHOOSE WHEN TO QUIT

```
message = ''

while message != 'quit':

    message = input('Enter a message: ')

    print(message)
```

USING A FLAG

```
message = ''
flag = True
while flag:
    message = input('Enter a number: ')
    if message == 'quit':
        flag = False
    else:
        print(message)
```

CHECK IF ALL NUMBERS ARE EVEN OR NOT

```
numbers = input('Enter the numbers separated by a space: ')
flag = True
for number in numbers.split(' '):
    if int(number)%2 != 0:
        flag = False
if flag:
    print('All numbers are even numbers')
else:
    print('All numbers are not even numbers')
```

USING BREAK TO EXIT A LOOP

- When we use break statement, the loop will terminate immediately.
- Applicable with for/while loop

```python
numbers = input('Enter the numbers separated by a space: ')

flag = True

for number in numbers.split(' '):

    if int(number)%2 != 0:

        flag = False

        break

if flag:

    print('All numbers are even numbers')

else:

    print('All numbers are not even numbers')
```

```
message = ''
while True:
    message = input('Enter the message: ')
    if message == 'quit':
        break
    else:
        print(message)
```

USING CONTINUE IN A LOOP

- Return to the beginning of the loop
- Print only the odd numbers

```
while True:
    number = input('Enter a number: ')
    if number == 'quit':
        break
    number = int(number)
    if number%2 == 0:
        continue
    print(number)
```

AVOIDING INFINITE WHILE LOOPS

- We can accidentally write while loops that runs infinitely

```
while True:
    print('This is an infinte while loop')
```

```
x = 1

while x <= 5:
    print(x)
    x += 1
```

USING WHILE WITH LISTS

- Do not use for loop when you're modifying the list inside it
- Python will find tough to track the elements
- Use while loop

GET A LIST OF NUMBERS FROM USERS

```
numbers = []
while True:
    number = input('Enter a number, type "quit" when done: ')
    if number == 'quit':
        break
    else:
        numbers.append(int(number))
print(numbers)
```

REMOVING ALL INSTANCES OF A VALUE FROM THE LIST

```
numbers = [1, 2, 1, 3, 1]

while 1 in numbers:

    numbers.remove(1)

print(numbers)
```

Thank You

